



Report Mini-project 2: Customer segmentation with clustering

Name: N'zebo Richard Anvo

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1. Introduction

In today's highly competitive e-commerce landscape, understanding customers and tailoring strategies to their needs is essential for sustaining growth and improving marketing effectiveness. Customer segmentation provides a systematic approach to group customers based on their characteristics and purchasing behaviours, enabling organizations to deliver personalized offers, optimize resource allocation, and strengthen customer relationships.

This project focuses on developing a robust customer segmentation model using a real-world e-commerce dataset. The study aims to leverage data exploration, preprocessing, feature engineering, and dimensionality reduction techniques to develop effective clustering models for customer segmentation. By uncovering meaningful customer groups, the e-commerce company can adopt a more customer-centric approach, leading to improved marketing efficiency, higher customer satisfaction, and ultimately, increased profitability.

2. Methods

The dataset is transnational, spanning five continents (Oceania, North America, Europe, Africa, and Asia) and consists of 951,668 transaction records collected between 1 January 2012 and 30 December 2016. [Table 1](#).

Table 1: Cluster Validation and Selection Methods

Method	Process	Strengths	Limitations
Elbow Method	Plot the Within-Cluster Sum of Squares (WCSS) for different cluster numbers; identify the "elbow" point where the reduction in WCSS slows.	Simple and intuitive; useful for estimating cluster range; widely applied in K-means clustering.	Subjective "elbow" point; may not give a clear result for complex datasets.
Silhouette Score	Compute average silhouette coefficient (cohesion vs. separation) for each cluster number; choose the value with highest score.	Provides a quantitative, objective measure; evaluates both compactness and separation.	Computationally expensive for large datasets; can be unstable with noisy data.
Hierarchical Clustering	Build a dendrogram by successively merging or splitting clusters based on similarity; cut the dendrogram at chosen distance threshold.	No need to pre-specify cluster number; visual representation of cluster relationships; captures nested structures.	Computationally intensive for large datasets; sensitive to noise and scaling.

2.1. The Elbow Method

The Elbow method was applied, which plots the WCSS against a range (2 -20) of potential k values. This range was chosen since extending it further did not affect the plot but increased computation time.

2.2. Silhouette Score Method

Silhouette score measures how similar an object is to its own cluster compared to other clusters. The method was applied for k in the range (2 to 20).

2.3. Hierarchical Clustering

The aggregate scale was used for the hierarchical clustering dendrogram. The number of clusters was limited to $p = 10$ due to computational limitations.

3. Results

3.1. Histogram observation

Frequency: right-skewed distribution, most customers have very low purchase frequencies, clustered near 0–500. Recency: Distribution is right-skewed, most customers have low recency values (3000–3500). CLV: Strong right skew, concentrated between (0–500). Avg_Unit_Cost: Heavily concentrated at 0–200, and most customers buy low-cost items. Age: Distribution is more uniform and peaks around ages 28–30, [Figure 1](#).

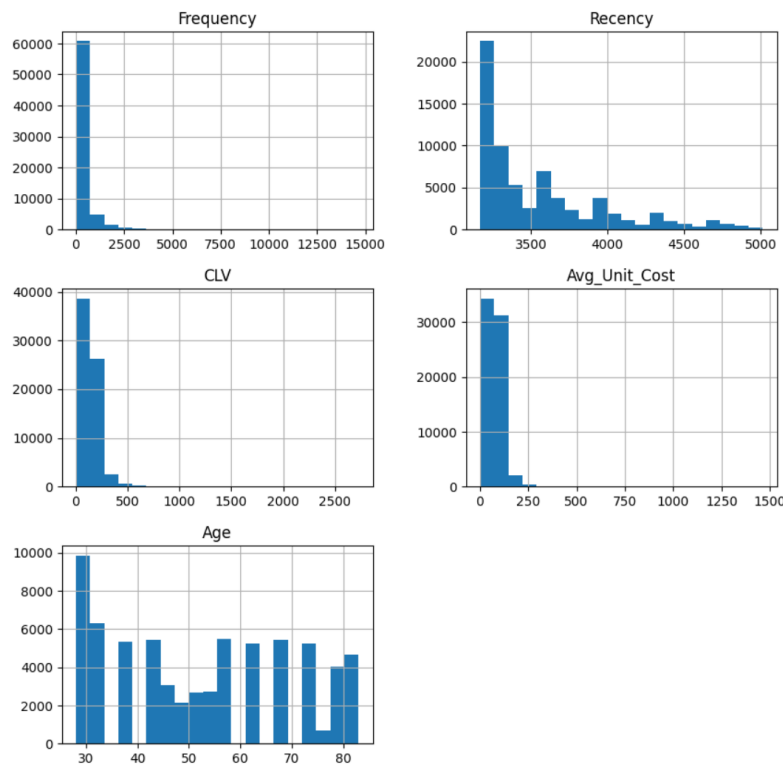


Figure 1: Feature histogram

3.2. Boxplots

The boxplots in Figure 2 reveal that most customer behaviours (frequency, CLV, and Av_Unit_cost) are concentrated at the lower end with many extremely high-value positive outliers, while recency shows moderate variability, and age is the most evenly distributed with minimal outliers.

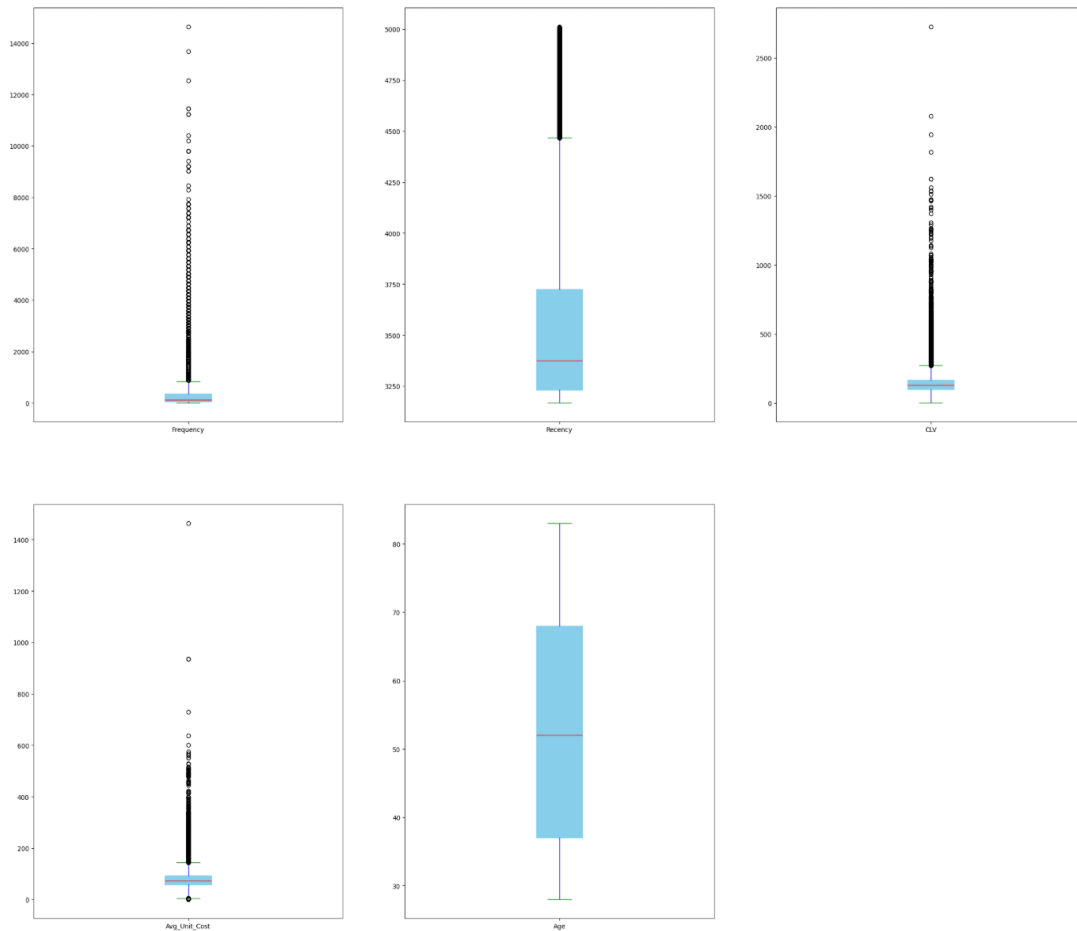


Figure 2: Boxplots for anomaly visualization

3.3. The Elbow Method

The Elbow Method was applied, which plots the WCSS against a range of potential k values. The WCSS decreases rapidly from k=1 to approximately k=4 and reveals a clear bend at around k=5 to k=6.

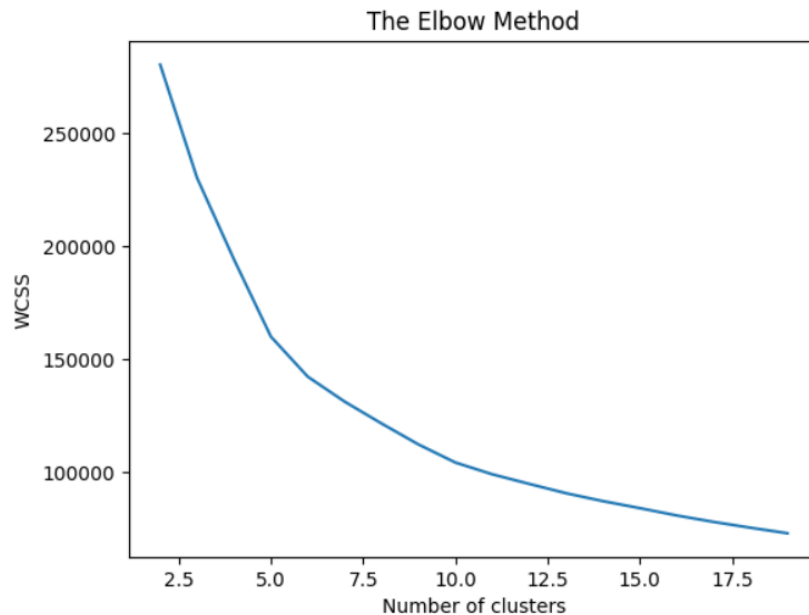


Figure 3: Elbow Method

3.4. Silhouette Score Method

The Silhouette score peaks at k=2 but shows a peak around k=4 to k=7. Beyond k=7, the score declines, [Table 2](#).

[Table 2](#): Silhouette score

Clusters	2	3	4	5	6	7	8	9	10
Average silhouette score	0.31662	0.23266	0.25472	0.27119	0.27374	0.27319	0.24444	0.25238	0.23494

3.5. Hierarchical Clustering

The hierarchical clustering dendrogram shows several potential cut points [Figure 4](#).

1. Scenario A: Clear split into two groups, long vertical line at the top, connecting two large branches, $k=2$.
2. Scenario B: Clear split into 3 groups, the line connects clusters within one and two main branches, $k=3$.
3. Scenario C: Clear split into 5 Groups. The vertical line is mixed with shorter and longer, $k=5$.

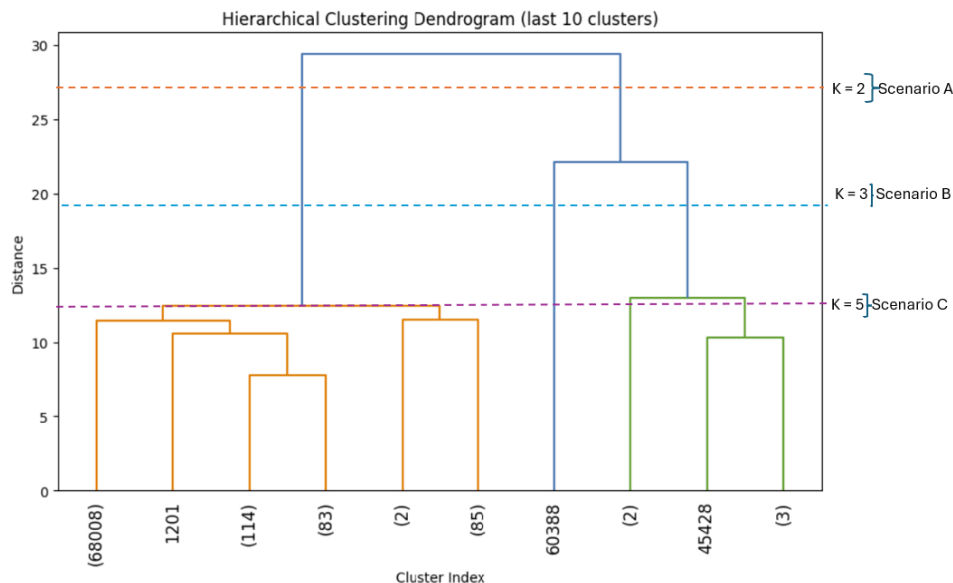


Figure 4: Hierarchical clustering dendrogram

3.6. Cluster selection

All three methods converge unequivocally on the same result. The evidence from computational efficiency (Elbow), cluster quality (Silhouette), and natural data structure (Dendrogram) is consistent; thus, the optimal number of clusters that best represents the data is $k = 5$.

3.6.1. Cluster boxplot

Figure 5 shows the clusters vs (frequency, recency, CLV, average unit cost, and customer age)

- The plot (Cluster vs Frequency) shows that customer frequency is extremely uneven. Cluster 1: represents the most frequent group, heavy buyers, clusters (0, 2) intermediate, purchase infrequently, clusters (3, 4): the least frequent purchase group. All outliers are in the positive direction.
- The plot (Cluster vs Recency) shows a clear separation of customer groups by recency, from very low recent buyers Cluster 1 to long-inactive customers Clusters (0, 2, 3, 4), highlighting where engagement and reactivation strategies should be focused.
- The CLV plot shows that Cluster 2 contains the most valuable customers (with extreme outliers driving revenue), while Clusters (0, 1) represent mid-value customers, and Clusters (3, 4) contain low-value buyers.
- The Average Unit Cost plot shows that Cluster 2 represents high-value customers, while Clusters (1, 3, 4) represent middle-market customers with moderate spending preferences, and Cluster 0 clearly identifies price-sensitive customers.
- The boxplot shows distinct age distributions across clusters. Cluster 4 is the youngest group (median 37 years). Clusters (1, 2) represent middle-aged to older adults (medians 47 and 53 years). Cluster (0, 3) is the oldest group (median 59 and 68 years), concentrated around 60–80 years.

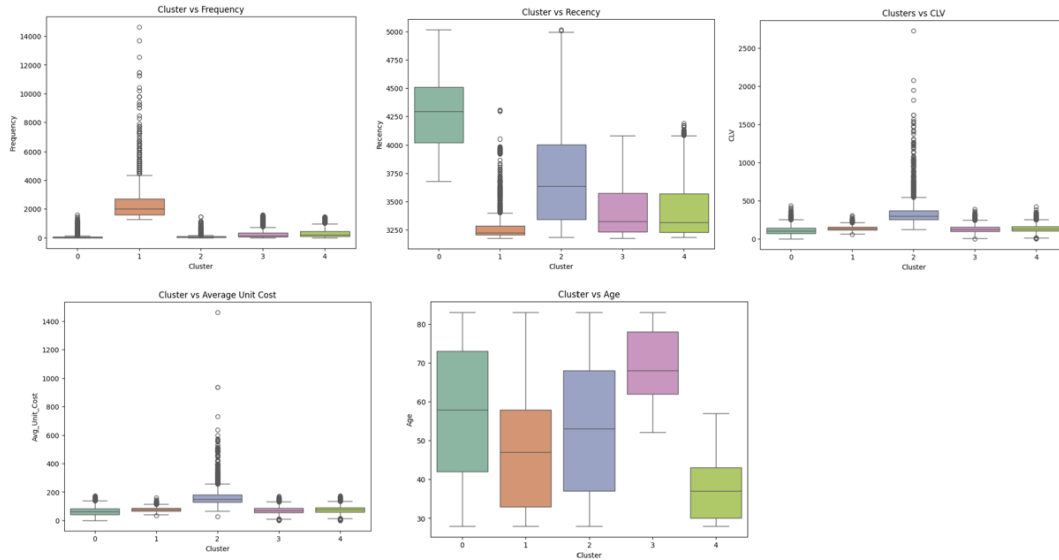


Figure 5: Boxplot

3.6.2. PCA and t-SNE

The PCA and t-SNE successfully reduce dimensionality while preserving cluster structure. Figure 6 demonstrates effective customer segmentation with clearly defined groups that represent distinct behavioral patterns.

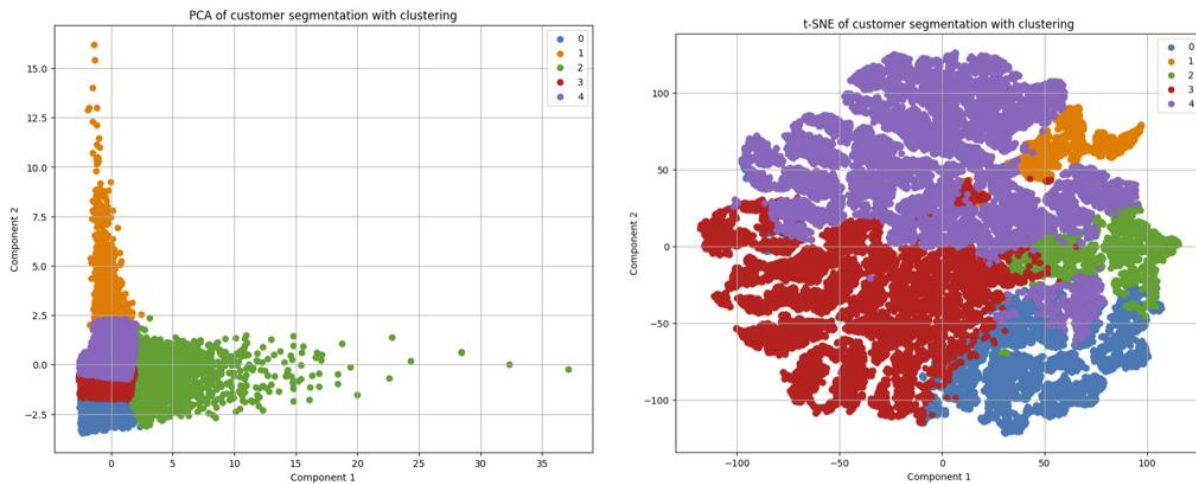


Figure 6: PCA, t-SNE of customer segmentation with clustering

4. Conclusion

Three methods Elbow Method, Hierarchical Dendrogram, and Silhouette Score, were applied to determine the optimal number of clusters. The Elbow Method revealed a clear bend at around $k=5$ to $k=6$. Similarly, the Hierarchical Clustering Dendrogram shows that a cut at $k=5$ offers a balanced solution. The Silhouette Score provided additional validation. While the highest score appeared at $k=2$ (0.317), this would group customers too broadly for actionable insights. Scores for $k=5$ (0.271) and $k=6$ (0.274) demonstrated more meaningful separation, balancing interpretability and cluster quality.

K-Means clustering was applied to partition customers into five distinct groups. PCA and t-SNE were employed for a more intuitive visualization of cluster separability, confirming the robustness of the chosen segmentation method.

This enables the business to identify distinct customer groups, such as loyal high-value buyers, occasional purchasers, and price-sensitive customers. With this level of segmentation, the company can design personalized offers, loyalty programs, and targeted campaigns to maximize customer value and long-term growth.